

Vivekanand Education Society's Institute of Technology



Department of Computer Engineering

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Project Synopsis Template (2025-26) - Sem VII

OralSure - AI Powered Self Assessment Tool for Oral Health

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Abstract

Oral health is a key indicator of general well-being, yet access to early diagnosis remains limited for many individuals. This project aims to develop a smartphone-based multilingual application powered by artificial intelligence that allows users to self-assess their oral health conditions focusing on both hard and soft tissues, along with simple symptom-based questions. The application will use deep learning-based image classification and grade identification of oral lesions to analyze uploaded images and identify possible oral health issues such as dental caries, ulcers, and gum inflammation. The app aims to increase awareness, encourage early detection, and promote timely professional consultation by offering a fast, accessible, and easy-to-use diagnostic tool integrated with an Oral Electronic Health Record (Oral EHR) system.

Introduction

Oral hygiene is a fundamental aspect of overall health, influencing not only the condition of teeth and gums but also contributing to systemic wellness. Maintaining good oral health supports essential functions such as eating, speaking, and social interaction, while also helping to prevent a range of dental and medical conditions. Regular oral care routines and early detection of issues play a critical role in reducing the risk of complications, improving quality of life, and promoting long-term health outcomes.

As technology continues to evolve, new opportunities are emerging to support oral hygiene through digital innovation. The widespread availability of smartphones, combined with the growing capabilities of artificial intelligence, opens the door to personalized and accessible tools for health monitoring. In this context, this project introduces a mobile application that uses AI-driven image analysis to assess intraoral selfies alongside brief symptom-based inputs. The app aims to provide users with preliminary insights into their oral health, promote early awareness, and encourage proactive care habits as a complement to professional dental services.

Problem Statement

The lack of accessible, multilingual tools for early diagnosis of hard and soft tissue oral conditions calls for an AI-powered mobile app combining image analysis, symptom-based assessment, and Oral EHR integration to enable timely detection and professional follow-up.

Proposed Solution

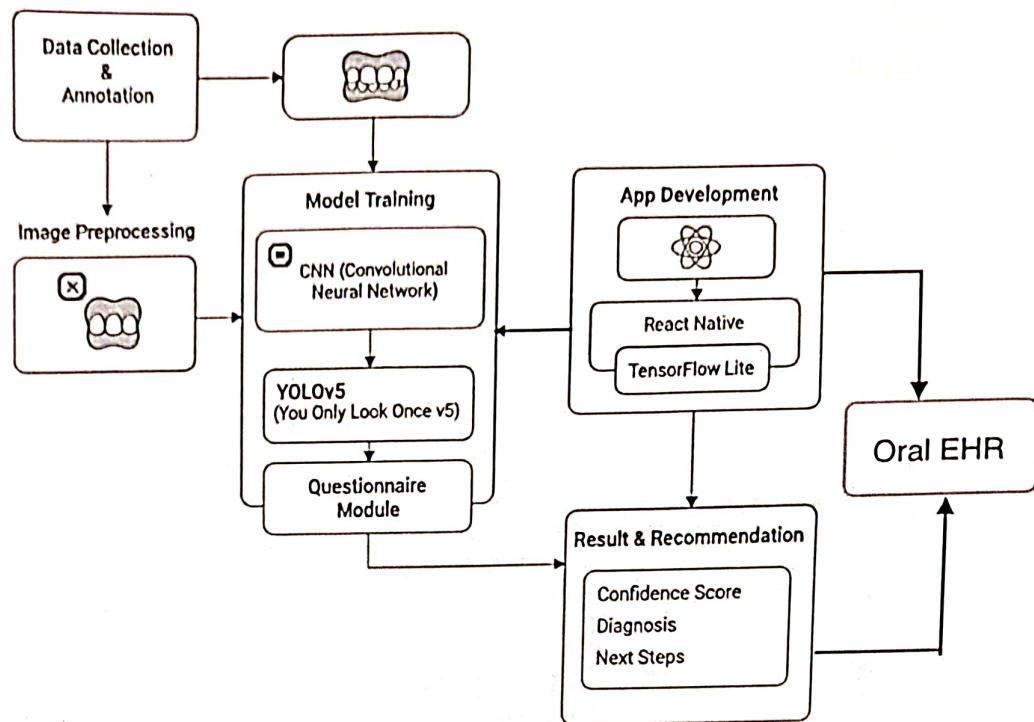
We propose building an AI-powered mobile application that enables users to assess their oral health condition quickly and conveniently. Users will be able to capture oral selfies using their smartphone camera and answer a short set of health-related questions within the app. Based on this input, the application will provide a preliminary oral health assessment.

To achieve this, the app will incorporate a lightweight Convolutional Neural Network (CNN) model trained to classify and detect various oral conditions from the uploaded images. The CNN will be selected and optimized specifically for mobile deployment to ensure that it performs efficiently and accurately in real time, even on devices with limited computing resources.

For enhanced performance, the model will be deployed using TensorFlow Lite or similar mobile-friendly machine learning frameworks. This ensures that the application remains responsive, fast, and user-friendly, without compromising the accuracy of the assessments.

The results provided to users will include a classification of the detected condition, relevant risk indicators, and actionable recommendations. These outputs are designed to help users make more informed decisions about whether or not to seek professional dental care.

Methodology / Block Diagram



- 1. Data Collection & Annotation:**
Collect clinical images of various oral conditions and label them with expert input.
- 2. Image Preprocessing:**
Enhance images using OpenCV (adjust contrast, brightness, crop relevant areas).
- 3. Model Training:**
 - Use a lightweight CNN (e.g., MobileNetV2) to classify oral conditions.
 - Apply YOLOv5 for detecting and highlighting affected regions.
- 4. Questionnaire Module:**
Gather user input on symptoms like pain, bleeding, or sensitivity to support diagnosis.
- 5. App Development:**
Build the app using React Native or Flutter and integrate AI models via TensorFlow Lite for offline use.
- 6. Result & Recommendation:**
Generate a diagnosis with confidence score and suggested next steps based on AI output and user input.

Hardware, Software, and Tools Requirements

Hardware:

- Smartphone with a camera (minimum 8MP recommended)
- Optional: Server for cloud-based processing if needed

Software and Tools:

- Python (for model training)
- TensorFlow / Keras (deep learning)
- OpenCV (image preprocessing)
- React Native (mobile app)
- Firebase / Node.js (backend and authentication)

Proposed Evaluation Measures

Metric	Description
Classification Accuracy	Measures how accurately the model classifies oral health conditions affecting hard and soft tissues.
Precision & Recall	Evaluates the model's ability to correctly identify conditions while minimizing false positives and false negatives.
Localization Accuracy (mAP / IoU)	Assesses how well the model detects and marks affected regions in oral cavity images.
On-Device Latency	Measures the time taken to process and return results on a smartphone for real-time use.
User Satisfaction	Assesses usability, clarity of results, and overall experience through user feedback and surveys.

Conclusion

In conclusion, this project seeks to revolutionize access to oral healthcare by harnessing the capabilities of artificial intelligence and mobile technology. Through a user-friendly mobile application, individuals will be able to perform quick and convenient self-assessments of their oral health by capturing intraoral selfies and responding to a short series of symptom-based questions. The core of the application will be a lightweight, mobile-optimized Convolutional Neural Network (CNN), deployed using frameworks such as TensorFlow Lite to ensure efficient performance on a wide range of smartphones, including those with limited processing power.

By providing preliminary assessments of common conditions like dental caries, ulcers, and gum inflammation, along with risk indicators and tailored recommendations, the app encourages early detection and informed decision-making. Overall, the application promotes preventive care, empowers users to take control of their oral health, and contributes to improved long-term health outcomes through timely intervention and increased awareness.

References

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